

upay Smart Corporate Flow

Entity Relationship Diagram — Data Model Specification

Derived from the project proposals prepared by Nafisha Anzum Dipra | August 2026

1. Purpose & scope

Neither source proposal specifies a relational data model, so this document infers one from the functional description in both documents: corporate clients uploading payroll batches, a deterministic account-validation layer, an Isolation Forest anomaly-detection layer, Maker-Checker authorization, and predictive liquidity forecasting. It is intended as a technical companion to the two proposals below, not a replacement for them.

2. Source documents

Document	Source File	Summary
upay Smart Corporate Flow	Project Proposal, pp1.docx	Next-gen B2B bulk payout platform proposal framing the Core Validation Layer, AI Risk Shield, and Maker-Checker workflow.
upay Corporate Assist	Business Case Proposal, pp2.docx	Companion business case for the same platform concept, framed as a graduation thesis project with an added confidentiality/scope section.

3. Entity overview

Entity	Description
CORPORATE_CLIENT	The enterprise customer onboarded onto the platform (e.g. a company processing payroll or vendor payouts).
USER_ACCOUNT	A platform user tied to a client, distinguished by role: Maker (HR / Finance Officer who uploads batches) or Checker (Finance Director who authorizes them).
CENTRAL_WALLET	The client's pre-funded disbursement wallet that payout batches draw against and liquidity forecasts target.
UPAY_ACCOUNT	A registered upay account record, used to confirm a payee's phone number is real and active before any transfer is attempted.
EMPLOYEE	A payee associated with a client and linked to an upay account; carries the historical average used by the anomaly-detection model.
PAYOUT_BATCH	One bulk-disbursement run created from a single spreadsheet upload (e.g. a month's payroll).
BATCH_LINE_ITEM	A single payee row within a batch, carrying its account-validation status and AI anomaly score.
RISK_ALERT	A flagged anomaly on a line item that requires Checker review before the batch can be released.
AUTHORIZATION_LOG	The Checker's OTP-based sign-off event that authorizes a batch for transmission.
LIQUIDITY_FORECAST	A predicted funding requirement for a client's central wallet over a future period.

4. Entity relationship diagram

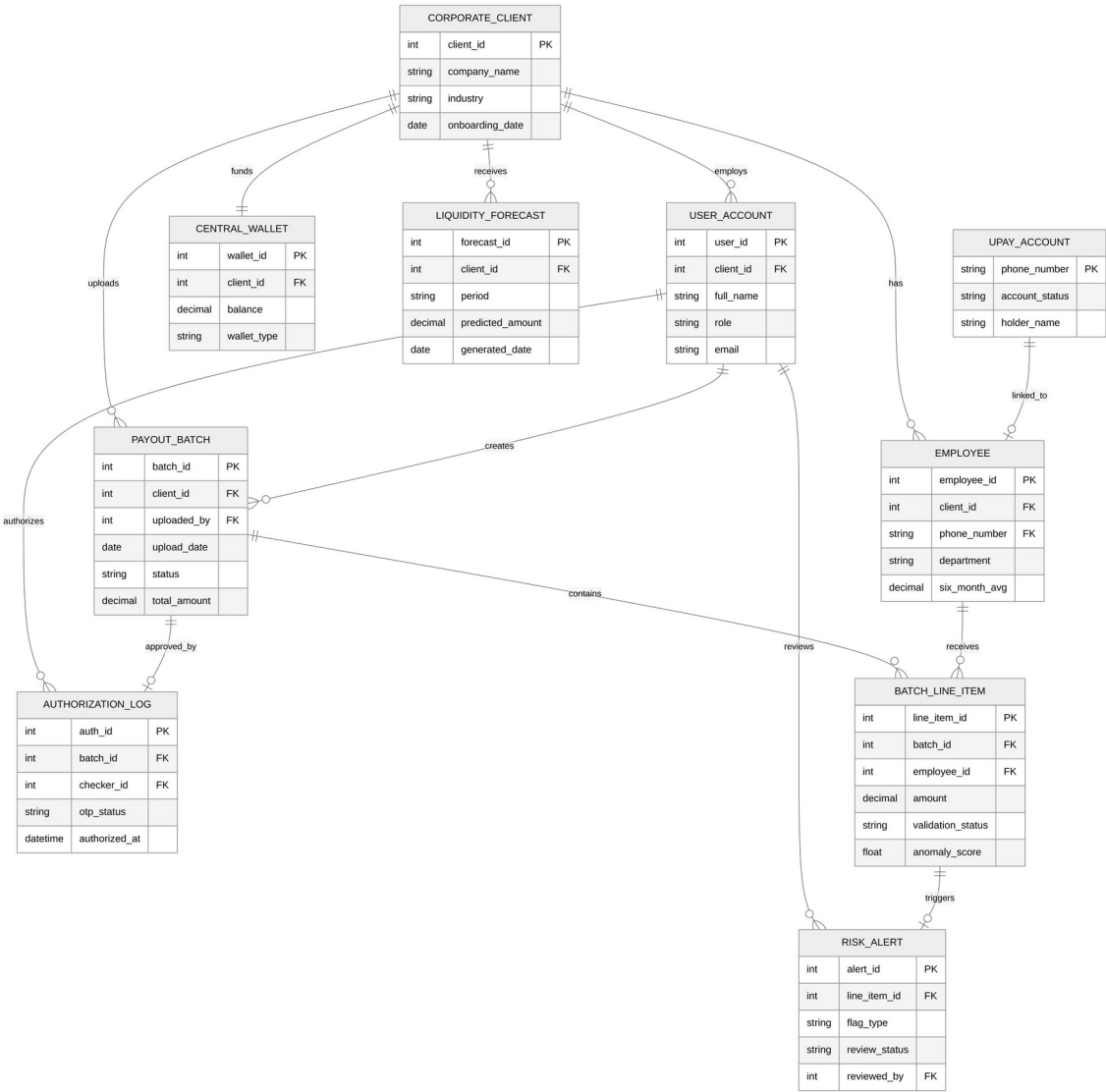


Figure 1. Proposed relational data model for upay Smart Corporate Flow / Corporate Assist.

5. Relationship summary

From	To	Cardinality	Relationship
CORPORATE_CLIENT	USER_ACCOUNT	1 : N	employs
CORPORATE_CLIENT	EMPLOYEE	1 : N	has
CORPORATE_CLIENT	CENTRAL_WALLET	1 : 1	funds
CORPORATE_CLIENT	PAYOUT_BATCH	1 : N	uploads
CORPORATE_CLIENT	LIQUIDITY_FORECAST	1 : N	receives
UPAY_ACCOUNT	EMPLOYEE	1 : 0..1	linked_to
USER_ACCOUNT	PAYOUT_BATCH	1 : N	creates (Maker)
USER_ACCOUNT	AUTHORIZATION_LOG	1 : N	authorizes (Checker)
USER_ACCOUNT	RISK_ALERT	1 : N	reviews (Checker)
PAYOUT_BATCH	BATCH_LINE_ITEM	1 : N	contains
PAYOUT_BATCH	AUTHORIZATION_LOG	1 : 0..1	approved_by
EMPLOYEE	BATCH_LINE_ITEM	1 : N	receives
BATCH_LINE_ITEM	RISK_ALERT	1 : 0..1	triggers

6. Key assumptions & notes

- This ERD is a proposed design, not a specification taken from the source documents — both proposals describe the platform's behavior but stop short of a data model.
- UPAY_ACCOUNT is modeled separately from EMPLOYEE because the platform validates phone numbers against upay's core account registry before an employee-linked payout row is trusted.
- BATCH_LINE_ITEM carries both the deterministic validation_status and the AI anomaly_score, since both checks apply per payee row within a batch.
- RISK_ALERT is split out from BATCH_LINE_ITEM so a flagged row keeps its own review trail (flag type, reviewer, status) separate from the transactional data.
- CENTRAL_WALLET is modeled as one wallet per client (1:1); if a client can hold multiple wallet types simultaneously, this should become 1:N.
- Historical payout averages for the anomaly model are assumed to be computed from past BATCH_LINE_ITEM rows for an employee, rather than stored as a separate table.